Nexus VIKOR

User Guide — Official Documentation

Organização: NEXUS-MCDM
Status: Distribuição Geral
Ambiente: Integrado / Standalone

1. Introduction to the Nexus MCDM Ecosystem

The **Nexus VIKOR** system belongs to the NEXUS-MCDM family of multi-criteria decision-making software. It delivers a state-of-the-art interface designed to guide users through structured decision modeling, ensuring mathematical consistency and ease of operation in both integrated and standalone installations.

2. Mathematical Formulation of Normalization

In MCDM, physical criteria values often come in heterogeneous units (e.g. currency, weight, efficiency, delay). Normalization is the procedure that converts these varying scales into a common, dimensionless range, typically [0, 1]. The Nexus system uses the following standard normalization formulations:

A. Linear Normalization for Benefit Criteria (Maximize):

For criteria where higher physical values represent better options (e.g. Quality, Profit, Throughput):
r_ij = (x_ij - min(x_j)) / (max(x_j) - min(x_j))
Where: *x_ij* is the physical consequence; *min(x_j)* is the worst performance; and *max(x_j)* is the best performance.

B. Linear Normalization for Cost Criteria (Minimize):

For criteria where lower physical values are desired (e.g. Cost, Lead Time, Risk):
r_ij = (max(x_j) - x_ij) / (max(x_j) - min(x_j))
Thus, the lowest cost achieves the highest utility score (1.0), and the highest cost maps to the lowest utility (0.0).

C. Vector Normalization:

Typically applied in distance-based MCDM solvers (such as TOPSIS), it divides each cell value by the column vector length:
r_ij = x_ij / sqrt(sum_{k=1}^m (x_kj^2))
This preserves the geometric proportion of raw data under vector transformation.

3. The 5-Step Decision Wizard

Building and resolving any decision problem follows 5 sequential steps:

Step 1: Rationality Definition

Select between **Compensatory** Rationality (where strong criteria performances can compensate for poor ones, e.g. AHP, TOPSIS, VIKOR) or **Non-Compensatory** Rationality (where veto thresholds strictly block trade-offs, e.g. ELECTRE, PROMETHEE).

Step 2: Decision Problematic

• **Choice (α):** Identify the best single option or a short elite list.
• **Ranking (β):** Classify all alternatives in order of preference (1st to last place).
• **Sorting (γ):** Assign alternatives to predefined categories.
• **Portfolio (δ):** Choose a combination of projects under a resource constraint (PROMETHEE V).

Step 3: MCDM Solver Selection

Pick the mathematical solver that fits the chosen rationality and problematic.

Step 4: Consequence Matrix Input

Enter the performance matrix. The application supports manual input and file imports from CSV, XLSX, and XLS in accordance with the TCC Consequence Matrix standard format.

Step 5: Preferences and Results

Configure criterion weights and local threshold curves. The engine then resolves the model, plots utilities, and enables download of the Complete Technical PDF Report.